

Bipolar Junction Transistor Analysis: Emitter Followers and Inverting Amplifiers

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Abstract

In this experiment, bipolar junction transistors were studied using two circuit configurations: the emitter follower and the common emitter amplifier. The emitter follower was used to examine impedance transformation, demonstrating a nearly constant output voltage and a low output resistance of approximately $14\,\Omega$. The common emitter amplifier was then used to study voltage amplification, where gain and signal inversion were recorded.

The frequency response of both circuits was analyzed. The emitter follower showed a gain approaching its peak at higher frequencies, while the common emitter amplifier exhibited reduced gain at low and high frequencies. Phase measurements confirmed the expected 180° inversion in the amplifier.

Overall, the experimental results were consistent with theoretical predictions and demonstrate how bipolar junction transistors can be used for both buffering and amplification in electronic systems.

1 Introduction

In this experiment, bipolar junction transistors (BJTs) were studied in two different circuit configurations: the emitter follower and the common emitter amplifier. These circuits demonstrate how transistors can be used to control and amplify electrical signals, as well as how their behavior depends on circuit design and biasing.

The emitter follower was first analyzed as a means of improving power delivery without providing voltage gain. Although the output voltage closely follows the input, this configuration significantly reduces the output resistance while maintaining a high input resistance. As a result, it acts as an impedance transformer, allowing a weak voltage source to effectively drive a load.

The emitter amplifier was then investigated as a voltage amplifying circuit. Unlike the emitter follower, this configuration provides substantial voltage gain and inverts the output signal. The experiment focused on understanding how gain depends on circuit components, as well as the role of proper biasing in maintaining operation in the active region.

Together, these circuits illustrate the relationship between gain, input resistance, and output resistance, and demonstrate how BJTs can be used for both buffering and amplification in realistic electronic systems.

2 Theoretical Framework

2.1 Transistor Operation

A BJT (npn type) consists of three terminals: the base, collector, and emitter. For normal operation in the active region, the base-emitter junction is forward biased while the collector-base junction is reverse biased. In this region, the collector current is proportional to the base current:

$$I_C \approx \beta I_B \quad (1)$$

where β is the current gain of the transistor. The currents are related by

$$I_E = I_C + I_B \quad (2)$$

and for large β ,

$$I_E \approx I_C. \quad (3)$$

The base-emitter voltage is approximately constant:

$$V_{BE} \approx 0.65 \text{ V} \quad (4)$$

so that

$$V_B \approx V_E + 0.65 \text{ V}. \quad (5)$$

2.2 Emitter Follower

In the emitter follower configuration, the output is taken from the emitter. The output voltage closely follows the input:

$$V_{out} \approx V_{in} - 0.65 \text{ V}. \quad (6)$$

This circuit has approximately unity voltage gain, but its main purpose is impedance transformation. The input resistance is

$$R_{in} = (\beta + 1)R_E \quad (7)$$

and the output resistance is

$$R_{out} = \frac{R_s}{\beta + 1}. \quad (8)$$

A large value of β therefore results in high input resistance and low output resistance, allowing more power to be delivered to a load.

2.3 Common Emitter Amplifier

In the common emitter configuration, the output is taken from the collector, resulting in both amplification and inversion of the signal. The small-signal voltage gain is given by

$$G = -\beta \frac{R_C}{R_E}. \quad (9)$$

A more accurate expression includes the intrinsic emitter resistance r_e :

$$G = -\beta \frac{R_C}{R_E + r_e} \quad (10)$$

where

$$r_e \approx \frac{25 \text{ mV}}{I_C}. \quad (11)$$

The negative sign indicates that the output is inverted relative to the input.

2.4 Biasing Considerations

To ensure proper operation, the transistor must be biased in the active region. For maximum output signal range, the collector voltage is typically set near half the supply voltage:

$$V_C \approx \frac{V_{CC}}{2}. \quad (12)$$

Using

$$V_C = V_{CC} - I_C R_C, \quad (13)$$

this gives

$$I_C \approx \frac{V_{CC}}{2R_C}. \quad (14)$$

The emitter and base voltages then follow as

$$V_E = I_E R_E, \quad V_B = V_E + 0.65 \text{ V}. \quad (15)$$

Correct selection of the biasing resistors ensures stable operation.

2.5 Circuit Comparison

The emitter follower and common emitter amplifier have different purposes. The emitter follower provides high input resistance and low output resistance with little to no voltage gain, making it more ideal for buffering. In contrast, the common emitter amplifier provides significant voltage gain but has a higher output resistance and inverts the signal.

3 Results and Analysis

3.1 Lab 6

DC Emitter Follower

The output voltage and current were measured for varying load resistances. The results are shown in Table 1.

R_L (Ω)	V_{out} (V)	I (A)
1000	10.81	0.01088
4700	10.91	0.00233
10000	10.92	0.00110
22000	10.93	0.00050
47000	10.86	0.00023
100000	10.93	0.00010

Table 1: Measured output voltage V_{out} and current for varying load resistance R_L .

The output voltage remains approximately constant near 10.9 V as the load resistance is varied over two orders of magnitude. In contrast, the current changes significantly. This indicates that the emitter follower behaves as a nearly ideal voltage source with low output resistance.

I-V Characteristics

The I-V characteristics of the emitter follower were obtained by plotting current as a function of output voltage.

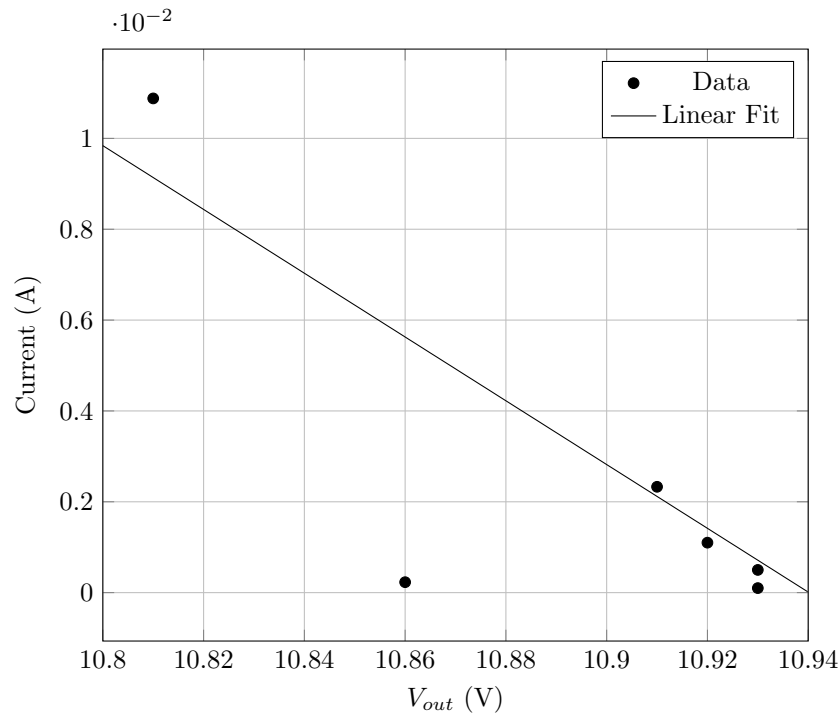


Figure 1: Current vs. output voltage for the emitter follower.

Output Resistance Determination

From the linear fit of the I-V data:

$$I = -0.0702V_{out} + 0.768 \quad (16)$$

The slope corresponds to:

$$\frac{dI}{dV} = -\frac{1}{R_{out}} \quad (17)$$

Thus,

$$R_{out} = \frac{1}{0.0702} \approx 14.2 \, \Omega \quad (18)$$

This small output resistance confirms that the emitter follower significantly improves the ability of the circuit to supply current to a load compared to the original voltage divider.

Power Delivery

The power delivered to the load is given by

$$P = IV \quad (19)$$

For $R_L = 1000 \, \Omega$:

$$P = (0.01088)(10.81) \approx 0.118 \, \text{W} \quad (20)$$

This shows that the circuit can deliver substantial power to low-resistance loads.

AC Frequency Response

The frequency response was measured by recording input and output voltages at various frequencies. The gain was calculated using

$$G = \frac{V_{out}}{V_{in}}. \quad (21)$$

f (Hz)	V_{in} (V)	V_{out} (V)	Gain
1000	1.38	0.834	0.604
2000	1.35	1.04	0.770
5000	1.30	1.11	0.854
10000	1.34	1.16	0.866
20000	1.34	1.17	0.873
50000	1.30	1.14	0.877
100000	1.33	1.17	0.880

Table 2: Measured gain at select frequencies.

Frequency Response (Gain vs. Frequency)

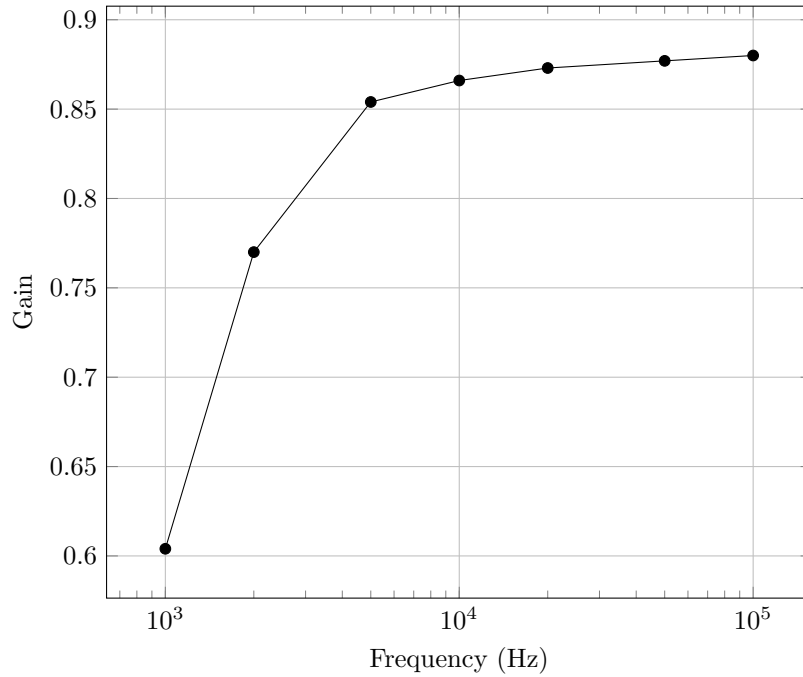


Figure 2: Gain as a function of frequency.

The gain increases with frequency and approaches unity at higher frequencies, as expected for an emitter follower. At low frequencies, the gain is reduced due to the effect of coupling capacitors, which introduce a high-pass filter behavior.

Discussion

The results confirm the expected behavior of the emitter follower. The circuit maintains a nearly constant output voltage while supplying varying current, indicating a low output resistance. The measured output resistance of approximately $14 \, \Omega$ agrees with theoretical expectations for a transistor-based buffer.

Additionally, the analysis shows that the circuit has a voltage gain close to its peak at higher frequencies, while lower frequencies are attenuated. Overall, the emitter follower functions effectively as

a buffer, providing impedance transformation and improved power delivery without significant voltage amplification.

3.2 Lab 7

Component Values and Biasing

Quantity	Value
R_C (theoretical)	10 k Ω
R_C (experimental)	10030 Ω
R_E (theoretical)	1 k Ω
R_E (experimental)	994 Ω
R_1 (theoretical)	10 k Ω
R_1 (experimental)	10 k Ω
R_2 (theoretical)	1 k Ω
R_2 (experimental)	993 Ω

Table 3: Theoretical and measured component values.

The measured values are in good agreement with the theoretical values. Small deviations can be caused by inherent uncertainty in the components.

Bias Voltages

Quantity	Value (V)
V_B (exp)	1.075
V_E (exp)	0.47
V_C (exp)	7.18
V_B (theory)	1.15
V_E (theory)	0.50
V_C (theory)	6.8

Table 4: Measured and theoretical bias voltages.

The measured values are close to the theoretical predictions. Small differences arise due to variations in transistor β and inherent component uncertainties.

Gain Calculation

Gain is defined as

$$G = \frac{V_{out}}{V_{in}}. \quad (22)$$

Example calculation at $f = 1000$ Hz:

$$G = \frac{2.95}{0.480} \approx 6.15 \quad (23)$$

This is consistent with the expected gain of approximately

$$G \approx \frac{R_C}{R_E} = \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega} = 10 \quad (24)$$

with reduction due to internal resistance and non-ideal behavior.

Bode Plot

The attenuation is defined as

$$A = 20 \log_{10}(G) \quad (25)$$

Example calculation at $f = 1000$ Hz:

$$A = 20 \log_{10}(6.15) \approx 15.77 \text{ dB} \quad (26)$$

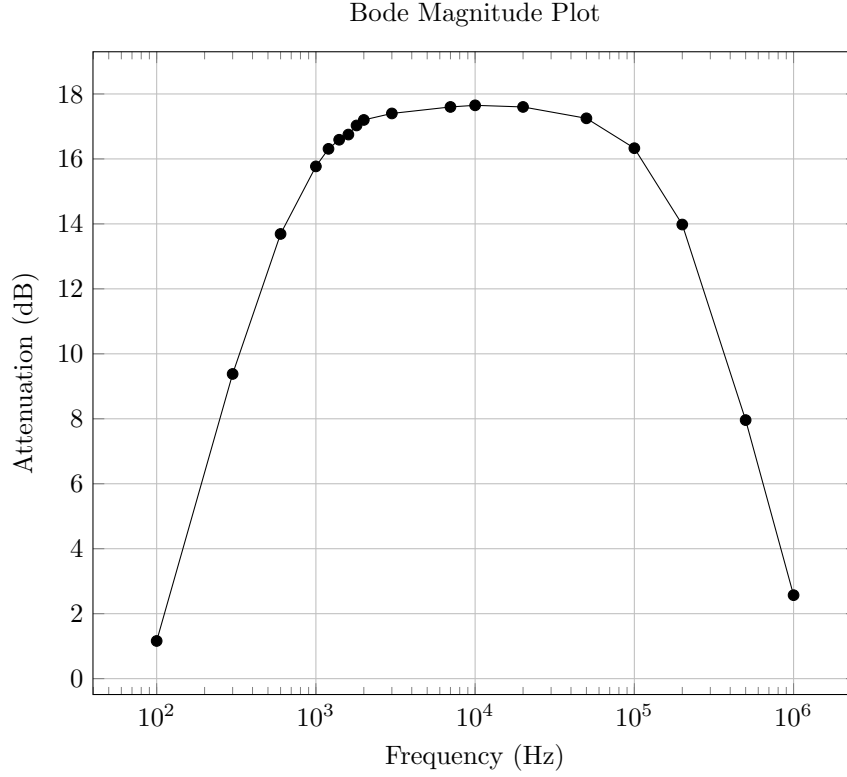


Figure 3: Attenuation (dB) as a function of frequency.

The gain increases at low frequencies, reaches a maximum in the mid-frequency range, and decreases at high frequencies. This behavior is characteristic of a band-limited amplifier.

Phase Shift

The phase shift was determined from the time delay using

$$\phi = \omega \Delta t \quad (27)$$

Example at $f = 1000$ Hz:

$$\phi = (6283)(0.000392) \approx 2.46 \text{ rad} \approx 141^\circ \quad (28)$$

This agrees with the measured value of approximately 144° .

Phase Shift vs Frequency

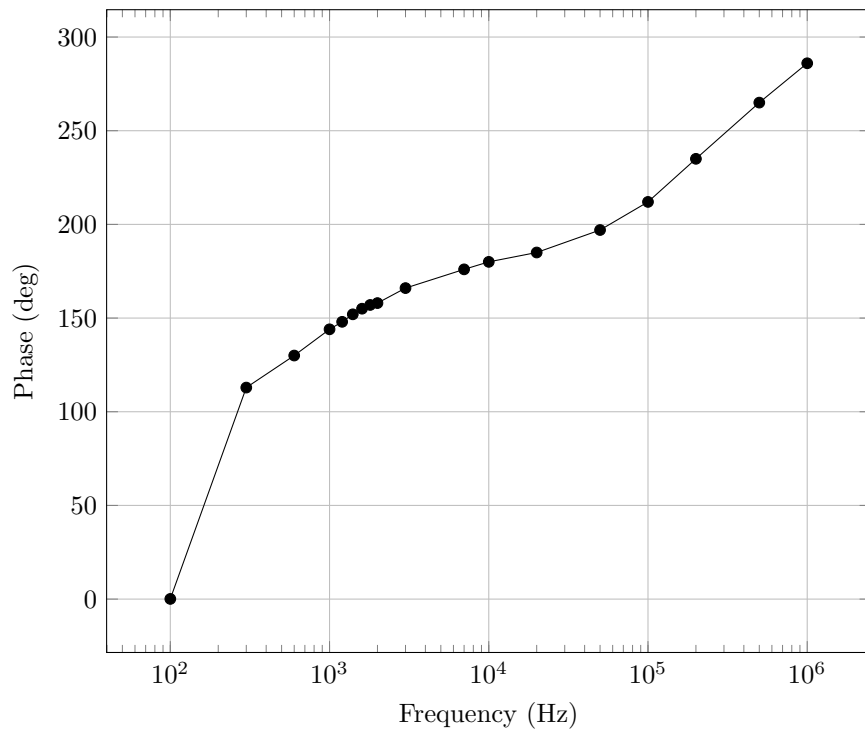


Figure 4: Phase shift as a function of frequency.

The phase shift approaches 180° in the mid-frequency range, confirming the expected inversion of the common emitter amplifier.

Discussion

The results show that the common emitter amplifier provides significant voltage gain with signal inversion. The measured gain is slightly lower than the theoretical value due to non-ideal effects and internal resistance.

The frequency response demonstrates typical amplifier behavior, with reduced gain at both low and high frequencies. At low frequencies, coupling capacitors limit the response, while at high frequencies, parasitic capacitances and transistor limitations reduce gain.

The phase shift measurements confirm that the output is inverted relative to the input, with the phase approaching 180° over most of the operating range.

Overall, the experimental results agree well with theoretical predictions and demonstrate the expected behavior of a common emitter amplifier.

4 Conclusions

In these experiments, the behavior of bipolar junction transistors was investigated using both the emitter follower and the emitter amplifier configurations. The results from both labs were consistent with theoretical expectations and demonstrated the distinct roles each circuit plays in electronic systems.

For the emitter follower, the circuit maintained a nearly constant output voltage while supplying a wide range of currents to the load. This resulted in a very small output resistance of approximately 14Ω , confirming that it is an impedance transformer.

For the common emitter amplifier, voltage gain and signal inversion were observed. The measured

gain was slightly lower than the theoretical value, which can be attributed to internal resistance and non-ideal behavior. The frequency response showed reduced gain at both low and high frequencies. The phase shift measurements further confirmed the expected inversion, with the phase approaching 180° over much of the range.

Overall, the experiments demonstrated how transistor circuits can be designed to achieve different objectives, such as impedance matching or voltage amplification.

References

- [1] *Lab 6: Bipolar Junction Transistors, Emitter Followers.*
- [2] *Lab 7: Bipolar Junction Transistor: Inverting Amplifiers.*
- [3] *Basic Electronics: An introduction to Electronics for Science Students*, Curtis A. Meyer

A Responses to questions in the manual

A.1 Lab 6: Preliminary lab questions

1. The circuit consists of two equal resistors ($4.7\text{ k}\Omega$) connected across a 12 V source. The Thevenin voltage is

$$V_{th} = V_{in} \frac{R_2}{R_1 + R_2} \quad (29)$$

Since $R_1 = R_2$,

$$V_{th} = 12 \cdot \frac{1}{2} = 6\text{ V} \quad (30)$$

The Thevenin resistance is found by shorting the voltage source:

$$R_{th} = R_1 \parallel R_2 = \frac{(4.7\text{ k})(4.7\text{ k})}{4.7\text{ k} + 4.7\text{ k}} = 2.35\text{ k}\Omega \quad (31)$$

So the solution is

$$V_{th} = 6\text{ V}, \quad R_{th} = 2.35\text{ k}\Omega$$

2. The input resistance seen at the base of the emitter follower is

$$R_{in} = (\beta + 1)R_E \quad (32)$$

Using $\beta \approx 100$ and $R_E = 470\text{ }\Omega$,

$$R_{in} = (101)(470) \approx 4.75 \times 10^4\text{ }\Omega = 47.5\text{ k}\Omega \quad (33)$$

The emitter follower shows a large input resistance.

3. Without the transistor, the voltage divider would be loaded by any additional resistance since $R_{th} = 2.35\text{ k}\Omega$ is relatively large. With the emitter follower, the effective load resistance becomes

$$R_{load,eff} \approx R_{in} \gg R_{th} \quad (34)$$

Since only a small amount of current is drawn, the output voltage remains roughly equal to the Thevenin voltage:

$$V_{out} \approx V_{th} \quad (35)$$

Therefore, the emitter follower prevents loading and stabilizes the voltage.

4. Looking into the output of the emitter follower, the Thevenin equivalent consists of a voltage source $V_{out} \approx V_B - 0.65$ and a small output resistance. The output resistance is

$$R_{out} = \frac{R_s}{\beta + 1} \quad (36)$$

where $R_s = R_{th} = 2.35 \text{ k}\Omega$. Thus,

$$R_{out} = \frac{2350}{101} \approx 23.3 \text{ }\Omega \quad (37)$$

Experimentally, R_{out} can be measured by first determining the open-circuit voltage V_{oc} , then attaching a known load R_L and measuring the new voltage V_L . The output resistance is then

$$R_{out} = R_L \left(\frac{V_{oc} - V_L}{V_L} \right) \quad (38)$$

A.2 Lab 6: In-lab questions

1. Replacing the voltage divider with its Thevenin equivalent gives $V_{th} = 6 \text{ V}$ and $R_{th} = 2.35 \text{ k}\Omega$. The effective load at the base is the input resistance of the emitter follower, given by

$$R_{in} = (\beta + 1)R_E \quad (39)$$

Using $\beta \approx 100$ and $R_E = 470 \text{ }\Omega$, we get

$$R_{in} \approx 47.5 \text{ k}\Omega \quad (40)$$

Since $R_{in} \gg R_{th}$, the voltage divider is minimally loaded.

2. The I-V curve for the voltage divider alone is given by its Thevenin equivalent:

$$I = \frac{V_{th} - V}{R_{th}} = \frac{6 - V}{2350} \quad (41)$$

This is a line with intercept $I = \frac{6}{2350}$ at $V = 0$ and slope $-\frac{1}{2350}$.

3. The emitter follower improves the voltage divider by reducing the effective output resistance. Without the transistor, the divider cannot supply significant current to small loads. With the emitter follower, the output behaves like a much better voltage source with low output resistance, allowing more power to be delivered to the load.
4. From the measured I-V curve of the emitter follower:

$$I = -0.0702V_{out} + 0.768 \quad (42)$$

The output resistance is

$$R_{out} = \frac{1}{0.0702} \approx 14.2 \text{ }\Omega \quad (43)$$

This is smaller than the voltage divider resistance of $2.35 \text{ k}\Omega$, therefore confirming the impedance transformation.

5. The power delivered to a load is

$$P = IV \quad (44)$$

For $R_L = 500 \Omega$, using $V \approx 10.9 \text{ V}$:

$$I = \frac{10.9}{500} = 0.0218 \text{ A} \quad (45)$$

$$P = (0.0218)(10.9) \approx 0.237 \text{ W} \quad (46)$$

Without the emitter follower, the voltage divider would drop significantly under this load, resulting in much lower power.

6. The circuit is not highly sensitive to the exact value of β because the input resistance is proportional to $(\beta + 1)$, which is already large. Variations in β do not notably change the overall behavior as long as $\beta \gg 1$.
7. If the voltage divider resistors were increased to $100 \text{ k}\Omega$, the Thevenin resistance would increase significantly:

$$R_{th} = 50 \text{ k}\Omega \quad (47)$$

This would make the source weaker, increasing sensitivity to loading and reducing performance, even with the emitter follower.

8. The minimum and maximum input voltages that can be followed are limited by the transistor's operating area. The output must satisfy the equation

$$V_{out} \approx V_{in} - 0.65 \quad (48)$$

The upper limit is constrained by the supply voltage (12 V), and the lower limit is constrained by the transistor turning off when V_{BE} is not forward biased. So the usable range is approximately:

$$0.65 \text{ V} < V_{in} < 12 \text{ V}$$

A.3 Lab 7: Preliminary lab questions

1. The base voltage is given by

$$V_B = \frac{1}{2} \frac{V_{CC}}{|G|} + 0.65 \quad (49)$$

Using $V_{CC} = 12 \text{ V}$ and $|G| = 10$:

$$V_B = \frac{1}{2} \cdot \frac{12}{10} + 0.65 = 0.6 + 0.65 = 1.25 \text{ V} \quad (50)$$

Thus, the base voltage should be approximately 1.25 V .

2. The voltage divider sets V_B such that

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2} \quad (51)$$

Solving for the ratio:

$$\frac{R_2}{R_1 + R_2} = \frac{1.25}{12} \approx 0.104 \quad (52)$$

Thus,

$$R_1 \approx 8.6R_2 \quad (53)$$

So the ratio $R_1 : R_2 \approx 8.6 : 1$.

3. The gain is given by

$$|G| \approx \frac{R_C}{R_E} \quad (54)$$

With $R_C = 4.7 \text{ k}\Omega$ and $|G| = 10$:

$$R_E = \frac{R_C}{|G|} = \frac{4700}{10} = 470 \Omega \quad (55)$$

The emitter resistance is

$$r_e \approx \frac{25 \text{ mV}}{I_C} \quad (56)$$

Using

$$I_C \approx \frac{V_{CC}}{2R_C} = \frac{12}{2 \cdot 4700} \approx 1.28 \text{ mA} \quad (57)$$

$$r_e \approx \frac{25 \text{ mV}}{1.28 \text{ mA}} \approx 19.5 \Omega \quad (58)$$

Since $r_e \ll R_E$, it does not significantly affect the gain.

4. The input resistance is

$$R_{in} = (\beta + 1)(R_E + r_e) \quad (59)$$

Using $\beta \approx 100$,

$$R_{in} \approx 101(470 + 20) \approx 49.5 \text{ k}\Omega \quad (60)$$

The Thevenin resistance of the divider should be much smaller to ensure stable biasing:

$$R_{th} \approx 1 \text{ k}\Omega \quad (61)$$

Choosing values consistent with the ratio $R_1 : R_2 \approx 8.6 : 1$, the choices are

$$R_1 = 10 \text{ k}\Omega, \quad R_2 = 1 \text{ k}\Omega$$

A.4 Lab 7: In-lab questions

1. The output voltage of the biasing divider without the transistor is

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2} = 12 \cdot \frac{1}{10 + 1} \approx 1.09 \text{ V} \quad (62)$$

After connecting the transistor, the measured value was $V_B = 1.075 \text{ V}$. The small decrease is due to base current loading the divider, but since the divider resistance is small, the effect is minimal.

2. The collector voltage was measured to be $V_C = 7.18$ V. The expected value was approximately 6.8 V. This is reasonably close and indicates that the transistor is operating in the active region.
3. The cutoff frequency for the input coupling capacitor is

$$f_c = \frac{1}{2\pi RC} \quad (63)$$

The relevant resistance is the input resistance seen by the capacitor:

$$R_{in} = (\beta + 1)(R_E + r_e) \quad (64)$$

Since R_{in} is large, a sufficiently large capacitor is required to ensure $f_c \leq 20$ Hz.

4. At $f \approx 1000$ Hz, the input and output signals were measured. The output signal is inverted relative to the input, consistent with the behavior of a common emitter amplifier.
5. The gain was measured as a function of frequency and plotted as a Bode plot. The -3 dB points correspond to where the attenuation drops by 3 dB from the maximum value. The high-frequency roll-off is primarily due to internal transistor capacitances rather than the oscilloscope input.
6. The gain was observed to be approximately constant for small input voltages, indicating linear operation. At larger input amplitudes, distortion occurs due to the transistor leaving the active region. The maximum undistorted input voltage is limited by the supply voltage and the biasing conditions.
7. The output resistance was measured by comparing the open-circuit voltage to the voltage across a known load resistor. The output resistance was found to be approximately equal to R_C , as expected for a common emitter amplifier.